B.Sc. II Semester (NEP)

DSC 2: Electricity and Magnetism

Unit 3 (11 hrs): Magnetism:

Statement of Biot-Savart law, Applications – (a) Magnetic field due to steady current in a long straight wire (b) Expression for force between two long straight conductors, Ampere's circuit law, Applications – (a) Magnetic field near a long wire carrying current (b) Magnetic field due to a solenoid, Current loop as a dipole, Torque on a dipole.

Electromagnetic induction:

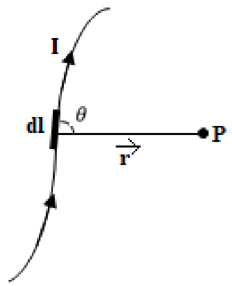
Review of Faradays laws of electromagnetic induction, Deduction of Faradays laws from Lorentz force. Induction and Relative Motion: A Conducting rod moving in a uniform magnetic field, Concept of self-induction and mutual induction, Energy stored in a magnetic field (derivation).

Magnetism

Biot- Savart law:

This is an empirical law which was formulated based of the experimental results and gives the relation for magnetic field produced by a current element.

Consider a current element of length dl carrying current I. P is a point at a distance r from the current element. Let θ be the angle between the current element and the line joining the point to the element.



The law states that the magnetic field dB at P due to current element is

1. Directly proportion to the strength of the current, I
2. Directly proportional to the length of the current element, dl
3. Directly proportional to the sine of the angle between the element and the line joining the point to the element, $sin\theta$
4. Inversely proportional to the square of the distance r of the point P from the element dl.

Combining all these factors

$$dB \propto \frac{I.dl.sin\theta}{r^2}$$

$$dB = \frac{\mu_0}{4\pi}.\frac{I.dl.sin\theta}{r^2}$$

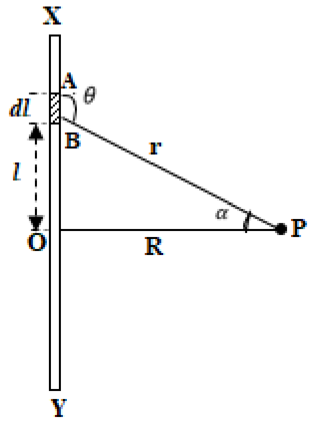
Where $\frac{\mu_0}{4\pi}$ is the proportionality constant and μ_0 is the permeability of free space.

The value of $\mu_0 = 4\pi \times 10^{-7} \frac{Wb}{Amp-m}$ $or\ H/m.$

In vector form Biot –Savart's is given by

$$dB = \frac{\mu_0}{4\pi}.\frac{I.\vec{dl} \times \vec{r}}{r^3}$$

Magnetic field due to steady current in a long straight wire:



Consider an infinitely long conductor placed in vacuum and carrying a current I amperes. Let P be a point at a distance R from the conductor. O is the foot of the perpendicular from P to the conductor. Consider a small element AB of length dl of conductor at a distance l from O. Let r be the distance of the element from the point P. Let θ be the angle between the current element and the line joining the point to the current element.

The magnitude of the field dB due to element AB at P is given by

$$dB = \frac{\mu_o}{4\pi} . \frac{I\, dl\, sin\theta}{r^2} \ldots \ldots \ldots . . (1)$$

The field due to the whole conductor is given by

$$B = \int_{-\infty}^{\infty} dB = \frac{\mu_o I}{4\pi} \int_{-\infty}^{\infty} \frac{sin\theta\, dl}{r^2} \ldots \ldots \ldots (2)$$

From figure,

$$r = (l^2 + R^2)^{1/2}$$

$$sin\theta = sin(\pi - \theta) = \frac{R}{r}$$

$$\therefore \quad sin\theta = \frac{R}{(l^2 + R^2)^{1/2}}$$

Substituting the value of r and $sin\theta$ in equation (2), we get,

$$B = \frac{\mu_o I}{4\pi} \int_{-\infty}^{\infty} \frac{R\, dl}{(l^2 + R^2)^{3/2}}$$

In order to evaluate the integral, consider

$$l = R\, tan\alpha$$

$$\therefore \quad dl = R\, sec^2\alpha\, d\alpha$$

$$\therefore \quad B = \frac{\mu_o I}{4\pi} \int_{-\pi/2}^{\pi/2} \frac{R\, R\, sec^2\alpha\, d\alpha}{(R^2\, tan^2\alpha + R^2)^{3/2}}$$

$$B = \frac{\mu_o I}{4\pi} \int_{-\pi/2}^{\pi/2} \frac{R^2\, sec^2\alpha\, d\alpha}{R^3 (1 + tan^2\alpha)^{3/2}}$$

$$B = \frac{\mu_o I}{4\pi} \int_{-\pi/2}^{\pi/2} \frac{d\alpha}{Rsec\, \alpha}$$

$$B = \frac{\mu_o I}{4\pi R} \int_{-\pi/2}^{\pi/2} cos\, \alpha\, d\alpha$$

$$B = \frac{\mu_o I}{4\pi R} \left[sin\, \alpha \right]_{-\pi/2}^{\pi/2}$$

$$B = \frac{\mu_o I}{4\pi R} (1 + 1)$$

$$B = \frac{\mu_o I}{2\pi R}$$

This is the expression for the magnetic field near a long straight conductor.

Expression for force between two long straight conductors:

Consider two infinitely long, straight parallel conductors X and Y separated by a distance r and carrying currents I_1 *and* I_2 respectively in the same direction as shown in figure. Magnetic fields are produced due to currents flowing in each conductor. Since each conductor is in the magnetic field produced by the other, each conductor experiences a force.

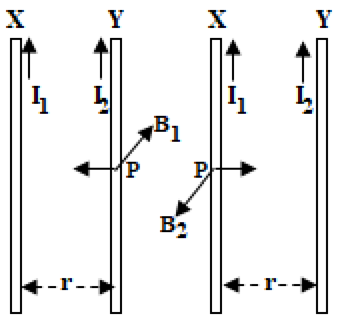


Fig (1) Fig (2)

Consider a point P on the conductor. The magnitude of the magnetic field at any point P on the conductor Y due to current I_1 in conductor X is

$$B_1 = \frac{\mu_o I_1}{2\pi r} \dots\dots\dots (1)$$

According to the right-hand rule (fore figure-current, middle figure-magnetic field, thumb-direction of force), the force on the conductor Y is inwards (Fig 1), that is towards conductor X. thus the magnitude of force experienced by length l of conductor Y is given by

$$F_2 = B_1\, I_2\, l \dots\dots\dots (2)$$

Substituting the value of B_1 from equation (1), we get

$$F_2 = \frac{\mu_o}{2\pi} . \frac{I_1 I_2 l}{r}$$

∴ Force per unit length of conductor,

$$F_2 = \frac{\mu_o}{2\pi} . \frac{I_1 I_2}{r} \dots\dots\dots (3)$$

Similarly, it can be shown that the conductor X also experiences a force and is directed towards the conductor Y as shown in Fig (2). The force on the conductor X per unit length is

$$F_1 = \frac{\mu_o}{2\pi} . \frac{I_1 I_2}{r} \dots\dots\dots (4)$$

Thus, the force between two long straight conductors is

$$F = \frac{\mu_o}{2\pi} . \frac{I_1 I_2}{r}$$

From the above expression it is clear that the parallel conductors carrying currents in the same direction attract each other, and parallel conductors carry

ing currents in opposite directions repel each other.

Ampere's circuit law:

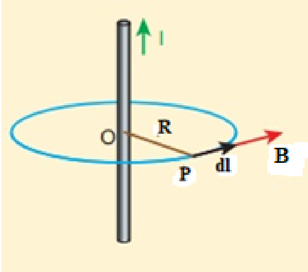
Oersted in 1820 discovered the existence of the magnetic field due to the current flowing in a conductor. Ampere derived a relationship between current I and the magnetic field B. The relationship is known as Ampere's law.

According to the law, "The line integral $\oint B\,dl$ for a closed curve is equal to μ_0 times the net current I through the area bounded by the curve". Thus

$$\oint B\,dl = \mu_0 I$$

Proof:

Consider a long straight conductor carrying a current I as shown in figure. Let the conductor be perpendicular to the plane of the paper, directed outwards. According to the Biot-Savart's law the magnitude of the magnetic induction at a distance R from the conductor is given by



$$B = \frac{\mu_0 I}{2\pi R} \dots\dots\dots (1)$$

At each point on this circle, B has a constant magnitude. Further, the direction of B at every point is along the tangent to the circle. That is, the angle between dl and B at every point on this path is zero. The line integral of magnetic field around this closed path is given by

$$\oint B\,dl\,cos\theta = \oint B\,dl = \frac{\mu_0 I}{2\pi R}\oint dl$$

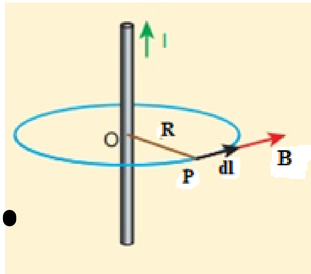
$$\oint B\,dl = \frac{\mu_0 I}{2\pi R}.2\pi R \qquad \because \oint dl = 2\pi R$$

$$\therefore \quad \oint B\,dl = \mu_0 I, \quad Hence\ the\ proof$$

<u>Applications of Ampere's circuit law</u>:

1. *Magnetic field due to a straight conductor carrying current:*

Consider a long straight conductor carrying current I in the direction shown in figure. The magnetic lines of force are concentric circles all centered at the conductor. Choose a circle of radius R as the closed path. It is desired to find the magnetic field B at a point P. the magnitude of B is the same everywhere on this closed path. The angle between B and dl is 0^0 everywhere on this path. Therefore, applying Ampere's law to this closed path, we have



$$\oint B\,dl = \mu_0 I$$

$$\oint B\,dl\,cos\,0 = \mu_0 I$$

$$B\oint dl = \mu_0 I$$

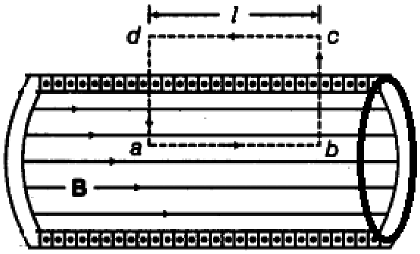
$$B.2\pi R = \mu_0 I$$

$$B = \frac{\mu_0 I}{2\pi R}$$

2. *Magnetic field due to a long solenoid carrying current:*

Consider a long air-cored solenoid having closely packed coils. Let I be the current flowing through the solenoid, n the number of turns per unit length. Choose a rectangular path abcd in a long solenoid as shown in figure. Let $ab = l$. The line integral of B over the closed path abcda is given by

$$\oint \vec{B}.\overrightarrow{dl} = \int_a^b \vec{B}.\overrightarrow{dl} + \int_b^c \vec{B}.\overrightarrow{dl} + \int_c^d \vec{B}.\overrightarrow{dl} + \int_d^a \vec{B}.\overrightarrow{dl}$$



$$\int_a^b \vec{B}.\overrightarrow{dl} = \int_a^b \vec{B}.\overrightarrow{dl}\, cos0 = B\int_a^b dl = Bl$$

$$\int_b^c \vec{B}.\overrightarrow{dl} = \int_d^a \vec{B}.\overrightarrow{dl} = 0 \quad \because angle\ between\ \vec{B}\ and\ \overrightarrow{dl}\ is\ 90^0$$

$$\int_c^d \vec{B}.\overrightarrow{dl} = 0 \quad \because Field\ outside\ the\ solenoid\ is\ zero.$$

$$\oint \vec{B}.\overrightarrow{dl} = Bl \ldots\ldots.(1)$$

According to Ampere's law we have

$$\oint \vec{B}.\overrightarrow{dl} = \mu_0 \times Current\ enclosed\ by\ a\ abcda$$

$$\oint \vec{B}.\overrightarrow{dl} = \mu_0 \times Number\ of\ turns\ in\ abcda \times I$$

$$\oint \vec{B}.\overrightarrow{dl} = \mu_0 \times nl \times I$$

$$\oint \vec{B}.\overrightarrow{dl} = \mu_0.n.l.I \ldots\ldots\ldots(2)$$

From equations (1) and (2), we have

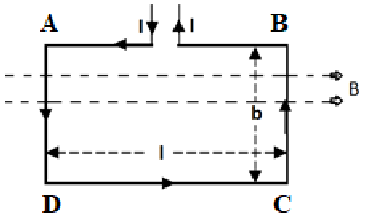
$$Bl = \mu_0.n.l.I$$

$$\therefore \quad B = \mu_0 nI$$

<u>Concept of dipole</u> & **<u>Current loop as a dipole</u>**:

According to the atomic theory, an atom consists of electrons rotating along closed orbital paths round the nucleus. The nucleus is a central heavy core consisting of neutrons and protons. The electron and the proton would appear as an electric dipole. The electrons which are revolving round the nucleus thus behave as a current loop. According to the explanation by Oersted, whenever current flows in a conductor, it gives rise to a magnetic field. Thus the current loop can be treated as a magnetic dipole. Further ampere explained that there are no such entities as individual independent magnetic north and south poles. He suggested that all observed magnetic phenomena are likely to be due to circulating currents, because of these circulating currents atoms and molecules can be treated as tiny magnets or magnetic dipoles.

Torque on a dipole:



Consider one turn of a rectangular coil $ABCD$ of length l and breadth b suspended in uniform magnetic field B which is parallel to the plane of the coil as shown in figure. Let a current I flow through the coil. Since the sides AB and DC are parallel to the magnetic field, the force experienced by them will be zero. The magnitude of the force on the side AD, is

$$F_{AD} = B\ I\ b\ sin\theta$$

$$F_{AD} = B\ I\ b$$

This force will be acting perpendicular to the plane of the coil and

Similarly, the magnitude of the force on the side BC is

$$F_{BC} = B\ I\ b$$

This force will be acting perpendicular to the plane of the coil and away from the observer. These two equal unlike forces constitute a couple which tends to rotate the coil about an axis through the suspension. The torque exerted by this couple in the coil, is

$$\tau = B\ I\ b \times l$$

$$\tau = B\ I\ A$$

$$\tau = M \times B$$

This is the expression for Torque on a dipole.

Electromagnetic induction:

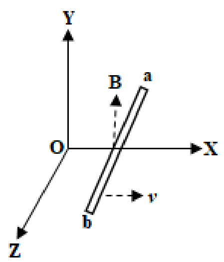
Faradays laws of electromagnetic induction:

I law: Whenever there is change in the magnetic flux linked with a circuit an emf and hence a current is induced in the circuit. The emf lasts as long as the change in flux persists.

II law: The magnitude of the induced emf is directly proportional to the rate of change of magnetic flux linked with the circuit. $e = -\frac{d\varphi}{dt}$

Deduction of Faradays laws from Lorentz force:

Consider a straight conducting rod ab lying with its length along the Z-axis in a uniform magnetic field $\vec{B}$ directed along the Y-axis and moving with a velocity $\vec{v}$ along the X-axis.



According to Lorentz force equation a charged particle on the conductor carrying a charge q will experience a force $\vec{F}$ given by

$$\vec{F} = q(\vec{v} \times \vec{B})$$

$$\frac{\vec{F}}{q} = (\vec{v} \times \vec{B})$$

But $\frac{\vec{F}}{q} = \vec{E}$, the induced electric field.

$$\therefore \quad \vec{E} = (\vec{v} \times \vec{B})$$

Consider an elemental length $\vec{dl}$ of the rod ab. The induced e.m.f. across the length $\vec{dl}$ is $\vec{E}.\vec{dl}$. Thus, the induced e.m.f. across the ends of the conductor due to the electric field is given by

$$e = \int_a^b \vec{E}.\vec{dl} = \int_a^b (\vec{v} \times \vec{B}).\vec{dl}$$

$$e = \int_a^b -\vec{B}.(\vec{v} \times \vec{dl})$$

$$e = -\vec{B}\int_a^b (\vec{v} \times \vec{dl}) \ldots\ldots(1) \qquad (A \times B).C = -B.(A \times C)$$

The conductor ab moves through a distance $\vec{v}\,dt$ along the X-axis in a time dt and in this time the area swept by the element $\vec{dl}$ is

$$\vec{v}\,dt \times \vec{dl} \qquad area = l \times b$$

Therefore the total area $\vec{ds}$ swept by the conductor ab in time dt is given by

$$\vec{ds} = \int_a^b (\vec{v}dt \times \vec{dl})$$

$$\frac{\vec{ds}}{dt} = \int_a^b (\vec{v} \times \vec{dl})$$

Substituting this value in eqn (1), we have

$$e = -\vec{B}\left[\frac{\vec{ds}}{dt}\right]$$

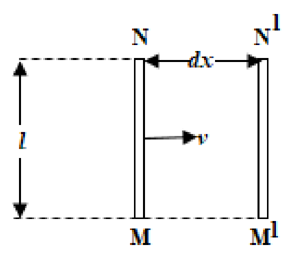
$$e = -\frac{\vec{B}.ds}{dt}$$

But, $\vec{B}.ds = d\phi$ a change in magnetic flux

$$\therefore \quad e = -\frac{d\phi}{dt}$$

This is the mathematical form of Faraday's II law of electromagnetic induction.

A Conducting rod moving in a uniform magnetic field:



Consider a straight conducting rod MN of length l moving at right angles to a uniform magnetic field B with a constant speed v. The magnetic field is perpendicular to the plane of the diagram and away from the observer as shown in figure.

Let the conductor move from MN to M^lN^l through a distance dx in time dt. Then area swept by the conductor $= MNM^lN^l = l\,dx$.

Therefore, the change in magnetic flux linked with the conductor in time dt is

$$d\phi = B(l\,dx)$$

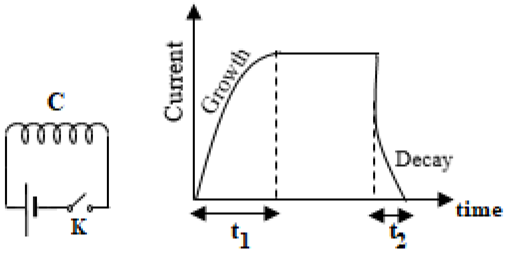
Magnitude of induced e.m.f. is

$$e = \left|\frac{d\phi}{dt}\right| = \left|Bl\frac{dx}{dt}\right| = B\,l\,v$$

$$\therefore\ \ motional\ emf, e = B\,l\,v$$

Concept of self-induction and mutual induction:

The phenomenon of self-induction was discovered by J.Henry in 1832. To understand the concept of self-inductance consider a coil C connected in series with a cell and a tap key as shown in figure. When the key is pressed the current in the circuit gradually increases and takes finite time t_1. As the current changes the magnetic flux linked with the coil also changes zero to a certain maximum value. In accordance with Faraday's I law an emf is induced in the coil. The emf induced opposes the growth of the current in the coil and is called the back emf. When the key is released the current in the coil decays from maximum to zero in time t_2. The flux linked with the coil also decreases from maximum to zero. This results in an induced emf but in the opposite direction. This emf is called the forward emf. It is observed that the time of growth t_1 is greater than the time of decay t_2 (fig). Hence the forward emf is greater than the back emf.



The phenomenon in which an emf is induced in a coil due to the change of current through the coil itself is known as self-induction. The property of a coil to oppose any change of current through it is known as self-inductance.

Coefficient of self-induction:

The magnetic flux ϕ linked with a coil at any instant is directly proportional to the current I through it at that instant. That is $\phi \propto I\ or\ \phi = LI$ where L is a constant called the self-inductance or coefficient of self-conduction. According to Faraday's II law

$$e = \frac{d\phi}{dt} = \frac{d}{dt}(LI) = L\frac{dI}{dt}$$

Hence, self-inductance of a coil is numerically equal to the emf induced in it when the rate of change of current through it is unity. The SI unit of self-inductance is henry.

Mutual induction: Consider a pair of coils placed side by side without touching each other. A varying current flowing through one coil induces an emf in the other. This phenomenon in which a change of current in one coil induces an emf in another neighbouring coil is called mutual induction.

The emf produced in the secondary coil is directly proportional to the rate of change of current through the primary coil.

$$e \propto \frac{dI}{dt} \quad or \quad e = M\frac{dI}{dt}$$

The constant M is called the coefficient of mutual induction or the mutual inductance between the coils.

Hence, mutual inductance between a pair of coils is numerically equal to the emf induced in the secondary coil when the current through the primary coil changes at unit rate.

<u>Energy stored in a magnetic field</u>:

Consider the circuit in which a source of emf E is connected to a resistance R and inductance L. From Kirchhoff's law,

$$E = IR + L\frac{dI}{dt}$$

$$EI = I^2R + LI\frac{dI}{dt}$$

The first term on R.H.S is the rate at which energy appears as heat in the resistance R. The energy that does not appear as heat must according to the rate of conservation of energy in the magnetic field. Therefore $LI\frac{dI}{dt}$ represents the rate at which the energy is stored in the magnetic field.

$$\frac{dU}{dt} = LI\frac{dI}{dt}$$

$$dU = LI\, dI$$

Therefore total energy stored in the magnetic field is

$$U = \int_0^{I_0} LI\, dI$$

$$U = \frac{1}{2}\, LI_0^2$$

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